

MINICRYPT WITH ALGEBRAIC STRUCTURE

ADVANCED TOPICS IN ~~CYBERSECURITY~~ CRYPTOGRAPHY (7CCSMATC)

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OUTLINE

Introduction

Computational Assumptions

(Bounded) Homomorphic One-way Functions (HOWFs)

(Bounded) Input-Homomorphic Weak Pseudorandom Functions (IHwPRFs)

INTRODUCTION

MAIN REFERENCE

Navid Alamati, Hart Montgomery, Sikhar Patranabis, and Arnab Roy. **Minicrypt Primitives with Algebraic Structure and Applications**. In: *Journal of Cryptology* 36.1 (Jan. 2023), p. 2.
DOI: 10.1007/s00145-022-09442-2



MINICRYPT: ONE-WAY FUNCTIONS ALL THE WAY DOWN

- OWFs $\Rightarrow$ PRGs [BM84]; PRGs $\Rightarrow$ PRFs [GGM86]; PRFs $\Rightarrow$ PRPs [LR88]
- OWFs $\Rightarrow$ Digital Signatures [Lam79; Mer79; NY89]

[AMPR23]

“It is natural to ask: is there any sort of mathematical structure that is inherent to [Cryptomania] primitives as well?”

MINICRYPT: ONE-WAY FUNCTIONS ALL THE WAY DOWN

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Question

Is it possible to construct Cryptomania primitives from simple Minicrypt primitives that are additionally equipped with some **algebraic structure**?

ALTERNATIVE PITCH I

Shor's Algorithm



Diffie-Hellman Land

Factorization Land

Lattice Land

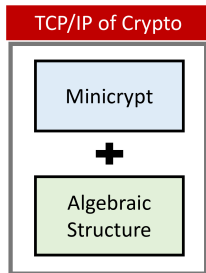
Code Land

Isogeny Land

“Post-Quantum”
Cryptomania

A “Narrow-Waist” Approach

Assumptions
• Discrete log • Search LWE • Square-root finding
• CDH • RSA
• DDH/MDDH/D-Lin • Decision LWE • QR/DCR • Approx. GCD • Hidden number problem • Finite Field isomorphism



Primitives
• Schnorr-style signatures • Trapdoor CRHFs • Succinct commitments
• Two-Party NIKE • CPA-Secure PKE • Trapdoor Functions • IBE • KDM-Secure PKE • DV-NIZK
• PIR • Lossy TDFs • OT and MPC

COMPUTATIONAL ASSUMPTIONS

DIFFIE-HELLMAN (DH)

Discrete Logarithms

Let p be a λ -bit prime and let g be a generator of the multiplicative group $\mathbb{Z}_p^*$. Given $g^a \bmod p$ find a .

DH

Let p be a λ -bit prime and let g be a generator of the multiplicative group $\mathbb{Z}_p^*$. Given

$$(g^a \bmod p, g^b \bmod p, u),$$

decide if $u \equiv g^{ab} \bmod p$ or random.

LEARNING WITH ERRORS (LWE)

Definition (LWE)

Let m, n, q be positive integers, χ be a probability distribution on $\mathbb{Z}$ and $\mathbf{s}$ be a uniformly random vector in $\mathbb{Z}_q^n$. Denote by $\mathcal{L}_{\mathbf{s}, \chi}$ the probability distribution on $\mathbb{Z}_q^n \times \mathbb{Z}_q^m$ obtained by choosing $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ uniformly at random, choosing every component of $\mathbf{e} \in \mathbb{Z}$ according to χ and returning

$$(\mathbf{A}, \mathbf{c}) = (\mathbf{A}, \mathbf{A} \cdot \mathbf{s} + \mathbf{e}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m.$$

Search-LWE recover $\mathbf{s}$ from $(\mathbf{A}, \mathbf{c}) = (\mathbf{A}, \mathbf{A} \cdot \mathbf{s} + \mathbf{e}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$ sampled according to $\mathcal{L}_{\mathbf{s}, \chi}$.

Decision-LWE decide if $(\mathbf{A}, \mathbf{c})$ is sampled from $\mathcal{L}_{\mathbf{s}, \chi}$ or the uniform distribution.

Search-LWE is like DLOG, Decision-LWE is like DH

DH AND LWE

DH Land	LWE Land
g	A
g^x	$A \cdot s + e$
$g^x \cdot g^y = g^{x+y}$	$(A \cdot s + e_0) + (A \cdot t + e_1) = A \cdot (s + t) + e'$
$(g^a)^b = (g^b)^a$	$t^T \cdot (A \cdot s + e) = t^T \cdot A \cdot s + t^T \cdot e$ $\approx t^T \cdot A \cdot s + e \cdot s$
(g, g^a, g^b, g^{ab}) $\approx_c (g, g^a, g^b, u)$	$(A, A \cdot s + e, t^T \cdot A + d^T, t^T \cdot A \cdot s + e')$ $\approx_c (A, A \cdot s + e, t^T \cdot A + d^T, u)$

(BOUNDED) HOMOMORPHIC ONE-WAY FUNCTIONS (HOWFs)

ONE-WAY FUNCTIONS (OWFs)

Definition (OWF)

A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is **one-way** if the following two conditions hold:

1. **(Easy to compute)** There exists a polynomial-time algorithm $\mathcal{A}_f(x)$ computing f , i.e. $\mathcal{A}_f(x) = f(x)$ for all x .
2. **(Hard to invert)** for every probabilistic polynomial-time $\mathcal{B}$, we have:

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{B}(1^n, f(x)) \in f^{-1}(f(x))] < 1/\text{poly}(n).$$

(BOUNDED) HOMOMORPHIC ONE-WAY FUNCTIONS (HOWFs)

Definition (Homomorphic OWF)

A homomorphic one-way function (HOWF) over an input group $(\mathcal{X}, \oplus)$ and an output group $(\mathcal{Y}, \otimes)$ is a one-way function $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that for any $x_0, x_1 \in \mathcal{X}$, we have $f(x_0 \oplus x_1) = f(x_0) \otimes f(x_1)$.

Definition (γ -Bounded Homomorphic OWF)

A γ -bounded homomorphic one-way function (γ -bounded HOWF) over an input group $(\mathcal{X}, \oplus)$ and an output group $(\mathcal{Y}, \otimes)$ is a one-way function $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$\mathcal{R} \left(f \left(\bigoplus_{j \in \{0, \dots, L-1\}} x_j \right) \right) = \mathcal{R} \left(\bigotimes_{j \in \{0, \dots, L-1\}} f(x_j) \right)$$

subject to the restriction $L \leq \gamma$ and some $\mathcal{R}$.¹

¹This definition differs from Def. 3.2 in [AMPR23]. Additionally, we should specify the groups as efficiently computable.

INSTANTIATION

From DLOG

$$\begin{aligned}f_g(x) &= g^x \\f_g(x_0 + x_1) &= g^{x_0} \cdot g^{x_1} \\&= f_g(x_0) \cdot f_g(x_1)\end{aligned}$$

From Search-LWE

$$\mathcal{R}(b \in \mathbb{Z}_q) = \begin{cases} 0 & |b| \leq q/4 \\ 1 & |b| > q/4 \end{cases}$$

$$f_A(s, e) = A \cdot s + e$$

$$\begin{aligned}\mathcal{R}(f_A(s + t, e + d)) &= \mathcal{R}(A \cdot s + e + A \cdot t + d) \\&= \mathcal{R}(f_A(s, e) + f_A(t, d))\end{aligned}$$

Here be dragons!

COLLISION RESISTANT HASH FUNCTIONS (FIRST ATTEMPT)

Definition (Collision-Resistant Hash Function)

A function $f : \{0, 1\}^{\text{poly}(n)} \rightarrow \{0, 1\}^n$ is called collision resistant if for every probabilistic polynomial-time $\mathcal{A}_f$, we have

$$\Pr [x_0 \neq x_1 \wedge f(x_0) = f(x_1) \mid x_0, x_1 \leftarrow \mathcal{A}_f()] < 1/\text{poly}(n).$$

- The definition is for an unkeyed function $f()$.
- Assume $\text{poly}(n) \geq n + 1$. Then there must exist x_0, x_1 such that $f(x_0) = f(x_1)$ by the Pigeon Hole principle.
- This means there **exists** some algorithm $\mathcal{A}_f$ that solves this problem for f : the algorithm that hardcodes x_0, x_1 .

COLLISION RESISTANT HASH FUNCTIONS (CORRECTED)

$$\text{Exp}_{H,\mathcal{A}}^{\text{cr}}(\lambda)$$

1 : $\text{hk} \leftarrow \$ H.\text{KGen}(1^\lambda)$

2 : $(x_0, x_1) \leftarrow \$ \mathcal{A}(1^\lambda, \text{hk})$

3 : **return** $H.\text{Eval}(\text{hk}, x_0) = H.\text{Eval}(\text{hk}, x_1) \wedge x_0 \neq x_1$

Definition (Collision-Resistant Hash Function)

An efficient keyed hash function $H = (H.\text{KGen}, H.\text{Eval})$ is called collision resistant (CR) if the advantage $\text{Adv}_{H,\mathcal{A}}^{\text{cr}}(\lambda)$ is negligible for all PPT algorithms $\mathcal{A}$. Here, the advantage is defined as

$$\text{Adv}_{H,\mathcal{A}}^{\text{cr}}(\lambda) := \Pr [\text{Exp}_{H,\mathcal{A}}^{\text{cr}}(\lambda)] \quad .$$

Setup:

1. Sample $2n$ uniform elements in $\mathcal{X}$:

$$\mathbf{X} := \begin{pmatrix} X_{0,0} & \dots & X_{0,n-1} \\ X_{1,0} & \dots & X_{1,n-1} \end{pmatrix} \leftarrow_{\$} \mathcal{X}^{2 \times n}$$

2. Compute

$$\mathbf{Y} := \begin{pmatrix} y_{0,0} & \dots & y_{0,n-1} \\ y_{1,0} & \dots & y_{1,n-1} \end{pmatrix} = \begin{pmatrix} f(X_{0,0}) & \dots & f(X_{0,n-1}) \\ f(X_{1,0}) & \dots & f(X_{1,n-1}) \end{pmatrix}$$

Call:

$$H_Y(\mathbf{r} = (r_0, \dots, r_{n-1})) = \bigotimes_{0 \leq j < n} y_{r_j, j}.$$

We assume there exists some adversary $\mathcal{A}$ that breaks collision resistance, i.e. outputs $\mathbf{r}, \mathbf{r}'$ s.t. $\mathbf{r} \neq \mathbf{r}'$ but $H_Y(\mathbf{r}) = H_Y(\mathbf{r}')$.

We construct an adversary $\mathcal{B}$ that breaks the one-wayness of $f(\cdot)$ using $\mathcal{A}$ as a subroutine.

1. $\mathcal{B}$ receives a challenge $y^* \in \mathcal{Y}$ and wants to find x^* s.t. $f(x^*) = y^*$.
2. $\mathcal{B}$ samples $2n$ uniform elements in $\mathcal{X}$: $\mathbf{X} := \{x_{b,j} \leftarrow \$ \mathcal{X}\}_{0 \leq j < n, b \in \{0,1\}}$ and sets $\mathbf{Y} := \{y_{b,j} := f(x_{b,j})\}_{0 \leq j < n, b \in \{0,1\}}$
3. $\mathcal{B}$ picks a random $i \in \{0, \dots, n-1\}$ and random $b^* \in \{0,1\}$ and sets $y_{b^*,i} := y^*$.

4. $\mathcal{B}$ runs $\mathcal{A}$ on $\mathbf{Y}$ and receives $\mathbf{r} \neq \mathbf{r}'$ back, s.t. $H_Y(\mathbf{r}) = H_Y(\mathbf{r}')$.
5. If $r_i = r'_i$ then $\mathcal{B}$ aborts (or outputs a random answer)

PROOF III

6. Assume $r_i = b^*$ (swap r_i and r'_i otherwise)

7. We have

$$\begin{aligned}
 & \bigotimes_{0 \leq j < n} y_{r_j, j} = \bigotimes_{0 \leq j < n} y_{r'_j, j} \\
 & \left(\bigotimes_{0 \leq j < i} y_{r_j, j} \right)^{-1} \cdot \bigotimes_{0 \leq j < n} y_{r_j, j} \cdot \left(\bigotimes_{i+1 \leq j < n} y_{r_j, j} \right)^{-1} = \left(\bigotimes_{0 \leq j < i} y_{r_j, j} \right)^{-1} \cdot \bigotimes_{0 \leq j < n} y_{r'_j, j} \cdot \left(\bigotimes_{i+1 \leq j < n} y_{r_j, j} \right)^{-1} \\
 & y^* = \left(\bigotimes_{0 \leq j < i} y_{r_j, j} \right)^{-1} \cdot \bigotimes_{0 \leq j < n} y_{r'_j, j} \cdot \left(\bigotimes_{i+1 \leq j < n} y_{r_j, j} \right)^{-1}
 \end{aligned}$$

8. Note that the RHS is independent of y^*

9. $\mathcal{B}$ outputs:

$$x^* = \left(\bigoplus_{0 \leq j < i} x_{r_j, j} \right)^{-1} \cdot \bigoplus_{0 \leq j < n} x_{r'_j, j} \cdot \left(\bigoplus_{i+1 \leq j < n} x_{r_j, j} \right)^{-1}$$

10. By the input-homomorphism, we have $f(x^*) = y^*$.

11. $\mathcal{B}$ succeeds whenever $\mathcal{A}$ succeeds and $r_i \neq r'_i$, which happens with probability at least $1/n$ for randomly chosen i .

SCHNORR-STYLE SIGNATURES

Let

- $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a HOWF
- n be some integer
- $H : \mathcal{Y} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function

Gen

- $\text{sk} := \mathbf{X} := \begin{pmatrix} X_{0,0} & \dots & X_{0,n-1} \\ X_{1,0} & \dots & X_{1,n-1} \end{pmatrix} \leftarrow_{\$} \mathcal{X}^{2 \times n};$
- $\text{vk} := \mathbf{Y} := \begin{pmatrix} f(X_{0,0}) & \dots & f(X_{0,n-1}) \\ f(X_{1,0}) & \dots & f(X_{1,n-1}) \end{pmatrix}$

Sign(sk, m)

1. $x^* \leftarrow_{\$} \mathcal{X}$ and set $y^* \leftarrow f(x^*)$
2. $\mathbf{r} := H(y^*, m)$
3. $\hat{x} := x^* \oplus \left(\bigoplus_{0 \leq j < n} x_{r_j, j} \right)^{-1}$

$$\sigma := (\mathbf{r}, \hat{x}, y^*)$$

Verify(vk, σ , m)

Check:

1. $\mathbf{r} \stackrel{?}{=} H(y^*, m)$
2. $y^* \stackrel{?}{=} f(\hat{x}) \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right)$

SCHNORR-STYLE SIGNATURES (DLOG INSTANTIATION: $f : x \rightarrow g^x \bmod p$)

Let

- n be some integer
- $H : \mathbb{Z}_p^* \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function

Gen

- $sk := \mathbf{X} := \begin{pmatrix} x_{0,0} & \dots & x_{0,n-1} \\ x_{1,0} & \dots & x_{1,n-1} \end{pmatrix} \leftarrow \mathbb{Z}_{p-1}^{2 \times n};$
- $vk := \mathbf{Y} := \begin{pmatrix} g^{x_{0,0}} & \dots & g^{x_{0,n-1}} \\ g^{x_{1,0}} & \dots & g^{x_{1,n-1}} \end{pmatrix}$

Sign(sk, m)

1. $x^* \leftarrow \mathbb{Z}_{p-1}$ and set $y^* \leftarrow g^{x^*} \bmod p$
2. $\mathbf{r} := H(y^*, m)$
3. $\hat{x} := x^* - \left(\sum_{0 \leq j < n} x_{r_j, j} \right)$

$$\sigma := (\mathbf{r}, \hat{x}, y^*)$$

Verify(vk, σ , m)

Check:

1. $\mathbf{r} \stackrel{?}{=} H(y^*, m)$
2. $y^* \stackrel{?}{=} g^{\hat{x}} \cdot \left(\prod_{0 \leq j < n} y_{r_j, j} \right) = g^{\hat{x} + \left(\sum_{0 \leq j < n} x_{r_j, j} \right)}$

ACTUAL SCHNORR SIGNATURES

Let

- $H : \mathbb{G} \times \{0, 1\}^* \rightarrow \mathbb{Z}_p$ be a hash function

Gen

- $sk := x \leftarrow \mathbb{Z}_p;$
- $vk := Y := g^x$

Claus-Peter Schnorr. **Efficient Identification and Signatures for Smart Cards**. In: *CRYPTO'89*. Ed. by Gilles Brassard. Vol. 435. LNCS. Springer, New York, Aug. 1990, pp. 239–252. DOI: 10.1007/0-387-34805-0_22

Sign(sk, m)

1. $x^* \leftarrow \mathbb{Z}_p$ and set $y^* \leftarrow g^{x^*}$
2. $r := H(y^*, m)$
3. $\hat{x} := x^* - x \cdot r$

$$\sigma := (r, \hat{x}, y^*)$$

Verify(vk, σ , m)

Check:

1. $r \stackrel{?}{=} H(y^*, m)$
2. $y^* \stackrel{?}{=} g^{\hat{x}} \cdot y^r = g^{\hat{x}} \cdot g^{xr}$

The proof of this (ubiquitous) construction relies on two proof techniques we have yet to cover:

- the Random Oracle Model
- Rewinding

Note 3.7. *The aforementioned signature scheme has an a priori bounded number of homomorphic operations, which allows it to be instantiated similarly using a γ -bounded HOWF, subject to the restriction that $2n < \gamma$. — [AMPR23]*

Here be dragons!

(BOUNDED) INPUT-HOMOMORPHIC
WEAK PSEUDORANDOM FUNCTIONS
(IHwPRFs)

STATISTICAL DISTANCE (SD) AKA TOTAL VARIATION DISTANCE

Definition (Statistical Distance)

Let $\mathcal{D}_0$ and $\mathcal{D}_1$ be two distributions with support $\mathcal{X}$. The Statistical Distance (SD) between $\mathcal{D}_0$ and $\mathcal{D}_1$ is:

$$\text{SD}(\mathcal{D}_0, \mathcal{D}_1) := 1/2 \cdot \sum_{x \in \mathcal{X}} |\mathcal{D}_0(x) - \mathcal{D}_1(x)|$$

Lemma (Distinguishing)

Given $\text{poly}(\lambda)$ samples, no adversary can distinguish $\mathcal{D}_0, \mathcal{D}_1$ with advantage $\geq 1/\text{poly}(\lambda)$ when $\text{SD}(\mathcal{D}_0, \mathcal{D}_1) < 1/\text{poly}(\lambda)$.

LEFTOVER HASH LEMMA (LHL)

Lemma (Leftover Hash Lemma)

Let $(\mathcal{X}, \oplus)$ be a finite group of size $|\mathcal{X}|$ and let n be a positive integer. Denote by $\mathcal{U}(\mathcal{X})$ the uniform distribution on $\mathcal{X}$. For any fixed $(2 \times n)$ -matrix of group elements $\mathbf{X} := \{x_{b,j}\}_{b \in \{0,1\}, 0 \leq j < n} \in \mathcal{X}^{2 \times n}$ let: $\mathcal{S}_{\mathbf{X}} = \left\{ \bigoplus_{0 \leq j < n} x_{j,r_j} : \mathbf{r} \leftarrow \{0,1\}^n \right\}$. For sufficiently large n it holds that

$$\Pr_{\mathbf{X} \leftarrow \mathcal{X}^{2 \times n}} \left[\text{SD}(\mathcal{S}_{\mathbf{X}}, \mathcal{U}(\mathcal{X})) > \sqrt[4]{\frac{|\mathcal{X}|}{2^n}} \right] \leq \sqrt[4]{\frac{|\mathcal{X}|}{2^n}}.$$

If $n > \log(|\mathcal{X}|) + \omega(\log(\lambda))$ and $\mathbf{X}$ uniform, then with overwhelming probability the statistical distance is negligible.

PRF (RECALL)

Definition (PRF)

A PRF is a keyed function $F_k : \mathcal{X} \rightarrow \mathcal{Y}$ with $k \leftarrow \$ \mathcal{K}$. We say F_k is (t, ε) -secure **PRF** if for Game_0 and Game_1 defined below we have:

$$\forall \mathcal{D} \in t \text{ steps: } \text{Adv}_F^{\text{prf}}(\mathcal{D}) = |\Pr[\mathcal{D}^{\text{Game}_1} = 1] - \Pr[\mathcal{D}^{\text{Game}_0} = 1]| < \varepsilon$$

Game_0	$F(x)$
1: $f \leftarrow \emptyset$	1: if $x \in \pi.\text{keys}$ then $y \leftarrow f[x]$
2: return $\mathcal{D}^f$	2: else
Game_1	3: $y \leftarrow \$ \mathcal{Y}; f[x] \leftarrow y$
1: $f \leftarrow \emptyset; k \leftarrow \$ \mathcal{K}$	4: $y \leftarrow F_k(x)$ // Game_1
2: return $\mathcal{D}^f$	5: return y

WEAK PRF

Definition (wPRF)

A PRF is a keyed function $F_k : \mathcal{X} \rightarrow \mathcal{Y}$ with $k \leftarrow \$ \mathcal{K}$. We say F_k is (t, ε) -secure **wPRF** if for Game_0 and Game_1 defined below we have:

$$\forall \mathcal{D} \in t \text{ steps: } \text{Adv}_F^{\text{wprf}}(\mathcal{D}) = |\Pr[\mathcal{D}^{\text{Game}_1} = 1] - \Pr[\mathcal{D}^{\text{Game}_0} = 1]| < \varepsilon$$

Game ₀	F()
1: $f \leftarrow \emptyset$	1: $x \leftarrow \$ \mathcal{X}$
2: return $\mathcal{D}^f$	2: if $x \in \pi.\text{keys}$ then $y \leftarrow f[x]$
Game ₁	3: else
1: $f \leftarrow \emptyset; k \leftarrow \$ \mathcal{K}$	4: $y \leftarrow \$ \mathcal{Y}; f[x] \leftarrow y$
2: return $\mathcal{D}^f$	5: $y \leftarrow F_k(x)$ // Game ₁
	6: return x, y

(BOUNDED) IHwPRF

Definition (Input-Homomorphic Weak PRF)

We call a keyed function $F_k : \mathcal{X} \rightarrow \mathcal{Y}$ an IHwPRF if the following conditions hold:

1. $F_k(\cdot)$ is a weak PRF.
2. Both $(\mathcal{X}, \oplus)$ and $(\mathcal{Y}, \otimes)$ are efficiently samplable groups.
3. The group operations $\oplus$ and $\otimes$, and their inverses are efficiently computable.
4. For every $k \in \mathcal{K}$ and every $x_0, x_1 \in \mathcal{X}$, we have:

$$F_k(x_0 \oplus x_1) = F_k(x_0) \otimes F_k(x_1).$$

γ -Bounded Variant

$$\mathcal{R} \left(F_k \left(\bigoplus_{0 \leq i < L} x_i \right) \right) = \mathcal{R} \left(\bigotimes_{0 \leq i < L} F_k(x_i) \right) \text{ for } L \leq \gamma \text{ and some } \mathcal{R}.$$

$$F_k(h) := h^k$$

BOUNDED IHwPRF FROM LWE

- Let $D_\chi()$ be an algorithm taking ℓ bits and outputting a sample following the distribution χ
- Let $F' : \mathbb{Z}_q^n \rightarrow \{0,1\}^\ell$ be a weak PRF outputting ℓ bits
- Define the bounded IHwPRF as:

$$F_{(k,s)}(\mathbf{a}) = \langle \mathbf{a}, \mathbf{s} \rangle + D_\chi(F'_k(\mathbf{a})).$$

- Define the relation as

$$\mathcal{R}(b \in \mathbb{Z}_q) = \begin{cases} 0 & |b| \leq q/4 \\ 1 & |b| > q/4 \end{cases}$$

Here be dragons!

OWFS FROM IHwUF/IHwPRF

Construction:

Let $F_k : \mathcal{X} \rightarrow \mathcal{Y}$ be an IHwPRF. Fix $n > 3 \log |\mathcal{X}|$ and sample

$$\mathbf{X} := \begin{pmatrix} x_{0,0} & \dots & x_{0,n-1} \\ x_{1,0} & \dots & x_{1,n-1} \end{pmatrix} \leftarrow \$ \mathcal{X}^{2 \times n}$$

Define $OWF_{\mathbf{X}} : \{0,1\}^n \rightarrow \mathcal{X}$ as

$$OWF_{\mathbf{X}}(\mathbf{r}) = \bigoplus_{0 \leq j < n} x_{r_j, j}.$$



The construction **never** calls $F_k()$!

We construct an adversary $\mathcal{B}$ against the IHwPRF security of F_k using a OWF adversary $\mathcal{A}$ against *OWF*.

1. $\mathcal{B}$ queries the wPRF oracle $2n$ times and obtains.

$$\mathbf{X} := \begin{pmatrix} x_{0,0} & \dots & x_{0,n-1} \\ x_{1,0} & \dots & x_{1,n-1} \end{pmatrix} \leftarrow_{\$} \mathcal{X}^{2 \times n} \text{ and } \mathbf{Y} := \begin{pmatrix} y_{0,0} & \dots & y_{0,n-1} \\ y_{1,0} & \dots & y_{1,n-1} \end{pmatrix}$$

2. It queries the oracle one more time to obtain x^*, y^* . It wishes to decide if $y_{b,j} = F_k(x_{b,j})$ and $y^* = F_k(x^*)$.
3. $\mathcal{B}$ sends $\mathbf{X}, x^*$ to the OWF adversary $\mathcal{A}$.
4. $\mathcal{A}$ outputs r , if $x^* \neq \bigoplus_{0 \leq j < n} x_{r_j, j}$ then $\mathcal{B}$ aborts.
5. $\mathcal{B}$ outputs $\bigotimes_{0 \leq j < n} y_{r_j, j} \stackrel{?}{=} y^*$.

6. If $y^*, y_{j,b}$ were sampled uniformly at random,
 - then $\Pr[y^* = \bigotimes_{0 \leq j < n} y_{r_j, j}] = 1/|\mathcal{Y}|$,
 - otherwise $\Pr[y^* = \bigotimes_{0 \leq j < n} y_{r_j, j}] = 0$.
7. By the LHL we know that for $\mathbf{X} \leftarrow \mathcal{X}^{2 \times n}$, $\mathbf{r} \leftarrow \{0, 1\}^n$, we have that $\bigoplus_{0 \leq j < n} x_{r_j, j}$ is statistically indistinguishable from x^* . Hence, $\mathcal{B}$ correctly simulates the one-wayness game for $\mathcal{B}$.
8. $\mathcal{B}$ succeeds roughly when $\mathcal{A}$ succeeds.

KeyGen:

1. Sample $k \leftarrow \mathcal{K}$
2. Sample $2n$ uniform elements in $\mathcal{X}$:

$$\mathbf{X} := \begin{pmatrix} x_{0,0} & \dots & x_{0,n-1} \\ x_{1,0} & \dots & x_{1,n-1} \end{pmatrix} \leftarrow \mathcal{X}^{2 \times n}$$

3. Compute

$$\mathbf{Y} := \begin{pmatrix} y_{0,0} & \dots & y_{0,n-1} \\ y_{1,0} & \dots & y_{1,n-1} \end{pmatrix} = \begin{pmatrix} F_k(x_{0,0}) & \dots & F_k(x_{0,n-1}) \\ F_k(x_{1,0}) & \dots & F_k(x_{1,n-1}) \end{pmatrix}$$

4. $sk := k, pk := (\mathbf{X}, \mathbf{Y})$

Enc:

1. Sample $\mathbf{r} \leftarrow \{0, 1\}^n$
2. Compute $c := \bigoplus_{0 \leq j < n} x_{r_j, j}$
3. Compute $e := m \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right)$
4. Output (c, e)

Dec:

1. Output $m' := e \otimes F_k(c)^{-1} = m \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right) \otimes F_k \left(\bigoplus_{0 \leq j < n} x_{r_j, j} \right)^{-1}$

Game ₀	$C(m_0, m_1)$
1 : $pk, sk \leftarrow \$ \text{KeyGen}(1^\lambda)$	1 : if $ m_0 \neq m_1 $ then
2 : $b \leftarrow \$ \{0, 1\}$	2 : return $\perp$
3 : $b' \leftarrow \mathcal{D}^c(pk)$	3 : $c \leftarrow \$ \text{Enc}(pk, m_b)$
4 : return $b = b'$	4 : return c

The definition of IND-CPA for PKE.

Game ₁	$C(m_0, m_1)$
1: $(X, Y), k \leftarrow \text{KeyGen}(1^\lambda)$	1: $\mathbf{r} \leftarrow \{0, 1\}^n$
2: $b \leftarrow \{0, 1\}$	2: $c \leftarrow \bigoplus_{0 \leq j < n} x_{r_j, j}$
3: $b' \leftarrow \mathcal{D}^c((X, Y))$	3: $e \leftarrow m_b \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right)$
4: return $b = b'$	4: return (c, e)

We simply instantiated our scheme.

Game ₂	$C(m_0, m_1)$
1: $(X, Y), k \leftarrow \text{KeyGen}(1^\lambda)$	1: $\mathbf{r} \leftarrow \{0, 1\}^n$
2: $b \leftarrow \{0, 1\}$	2: $c \leftarrow \bigoplus_{0 \leq j < n} x_{r_j, j}$
3: $b' \leftarrow \mathcal{D}^c((X, Y))$	3: $e \leftarrow m_b \otimes F_k(c)$
4: return $b = b'$	4: return (c, e)

Correctness of IHwPRF

Game ₃	$C(m_0, m_1)$
1: $(X, Y), k \leftarrow_{\$} (\mathcal{X}^{2 \times n}, \mathcal{Y}^{2 \times n}), \mathcal{K}$	1: $r \leftarrow_{\$} \{0, 1\}^n$
2: $b \leftarrow_{\$} \{0, 1\}$	2: $c \leftarrow \bigoplus_{0 \leq j < n} x_{r_j, j}$
3: $b' \leftarrow \mathcal{D}^C((X, Y))$	3: $e \leftarrow m_b \otimes F_k(c)$
4: return $b = b'$	4: return (c, e)

wPRF Security of $F_k()$

Game ₄	$C(m_0, m_1)$
1: $(X, Y), k \leftarrow_{\$} (\mathcal{X}^{2 \times n}, \mathcal{Y}^{2 \times n}), \mathcal{K}$	1: $\mathbf{r} \leftarrow_{\$} \{0, 1\}^n$
2: $b \leftarrow_{\$} \{0, 1\}$	2: $c \leftarrow_{\$} \mathcal{X}$
3: $b' \leftarrow \mathcal{D}^c((X, Y))$	3: $e \leftarrow m_b \otimes F_k(c)$
4: return $b = b'$	4: return (c, e)

LHL

Game ₅	$C(m_0, m_1)$
1: $(X, Y), k \leftarrow \$ (\mathcal{X}^{2 \times n}, \mathcal{Y}^{2 \times n}), \mathcal{K}$	1: $r \leftarrow \$ \{0, 1\}^n$
2: $b \leftarrow \$ \{0, 1\}$	2: $c \leftarrow \$ \mathcal{X}$
3: $b' \leftarrow \mathcal{D}^C((X, Y))$	3: $e \leftarrow \$ \mathcal{Y}$
4: return $b = b'$	4: return (c, e)

wPRF security of $F_k()$

KeyGen:

1. Sample $k \leftarrow \$ \mathcal{K}$
2. Sample $2n$ uniform elements in $\mathcal{X}$:

$$\mathbf{X} := \begin{pmatrix} X_{0,0} & \dots & X_{0,n-1} \\ X_{1,0} & \dots & X_{1,n-1} \end{pmatrix} \leftarrow \$ \mathcal{X}^{2 \times n}$$

3. Compute

$$\mathbf{Y} := \begin{pmatrix} F_k(X_{0,0}) & \dots & F_k(X_{0,n-1}) \\ F_k(X_{1,0}) & \dots & F_k(X_{1,n-1}) \end{pmatrix}$$

4. $sk := k, pk := (\mathbf{X}, \mathbf{Y})$

KeyGen:

1. Sample $k \leftarrow \$ \mathbb{Z}_{p-1}$
2. Sample $2n$ uniform elements in $\mathbb{Z}_p^*$:

$$\mathbf{X} := \begin{pmatrix} X_{0,0} & \dots & X_{0,n-1} \\ X_{1,0} & \dots & X_{1,n-1} \end{pmatrix} \leftarrow \$ (\mathbb{Z}_p^*)^{2 \times n}$$

3. Compute

$$\mathbf{Y} := \begin{pmatrix} X_{0,0}^k & \dots & X_{0,n-1}^k \\ X_{1,0}^k & \dots & X_{1,n-1}^k \end{pmatrix}$$

4. $sk := k, pk := (\mathbf{X}, \mathbf{Y})$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0, 1\}^n$
2. Compute $c := \bigoplus_{0 \leq j < n} x_{r_j, j}$
3. Compute $e := m \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right)$
4. Output (c, e)

Dec:

1. Output $m' := e \otimes F_k(c)^{-1}$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0, 1\}^n$
2. Compute $c := \prod_{0 \leq j < n} x_{r_j, j}$
3. Compute $e := m \cdot \left(\prod_{0 \leq j < n} y_{r_j, j} \right)$
4. Output (c, e)

Dec:

1. Output $m' := e \cdot c^{-k}$

Taher ElGamal. **A Public Key Cryptosystem and a Signature Scheme Based on Discrete Logarithms**. In: *CRYPTO'84*. Ed. by G. R. Blakley and David Chaum. Vol. 196. LNCS. Springer, Berlin, Heidelberg, Aug. 1984, pp. 10–18. DOI: [10.1007/3-540-39568-7_2](https://doi.org/10.1007/3-540-39568-7_2)

KeyGen:

1. Sample $k \leftarrow \$ \mathbb{Z}_{p-1}$
2. Sample $\mathbf{X} \leftarrow \$ (\mathbb{Z}_p^*)^{2 \times n}$
3. Compute $\mathbf{Y} := \{x_{b,j}^k\}_{b \in \{0,1\}, 0 \leq j < n}$
4. $\text{sk} := k, \text{pk} := (\mathbf{X}, \mathbf{Y})$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0,1\}^n$
2. Compute $\mathbf{c} := \prod_{0 \leq j < n} x_{r_j,j}$
3. Compute $e := m \cdot \left(\prod_{0 \leq j < n} y_{r_j,j} \right)$
4. Output $(\mathbf{c}, e)$

Dec:

1. Output $m' := e \cdot c^{-k}$

KeyGen:

1. Sample $k \leftarrow \$ \mathbb{Z}_{p-1}$
2. $x = g$
3. Compute $y = g^k$
4. $\text{sk} := k, \text{pk} := (x, y)$

Enc:

1. Sample $r \leftarrow \$ \mathbb{Z}_{p-1}$
2. Compute $c := g^r$
3. Compute $e := m \cdot y^r$
4. Output (c, e)

Dec:

1. Output $m' := e \cdot c^{-k}$

Oded Regev. **On lattices, learning with errors, random linear codes, and cryptography**. In: *Journal of the ACM* 56.6 (Sept. 2009), 34:1–34:40. ISSN: 0004-5411 (print), 1557-735X (electronic). DOI: <http://doi.acm.org/10.1145/1568318.1568324>

Ron Rothblum. **Homomorphic Encryption: From Private-Key to Public-Key**. In: *TCC 2011*. Ed. by Yuval Ishai. Vol. 6597. LNCS. Springer, Berlin, Heidelberg, Mar. 2011, pp. 219–234. DOI: [10.1007/978-3-642-19571-6_14](https://doi.org/10.1007/978-3-642-19571-6_14)

KeyGen:

1. Sample $k \leftarrow \mathcal{K}$
2. Sample $2n$ uniform elements in $\mathcal{X}$:

$$\mathbf{X} := \begin{pmatrix} X_{0,0} & \dots & X_{0,n-1} \\ X_{1,0} & \dots & X_{1,n-1} \end{pmatrix} \leftarrow \mathcal{X}^{2 \times n}$$

3. Compute

$$\mathbf{Y} := \begin{pmatrix} F_k(X_{0,0}) & \dots & F_k(X_{0,n-1}) \\ F_k(X_{1,0}) & \dots & F_k(X_{1,n-1}) \end{pmatrix}$$

4. $\text{sk} := k, \text{pk} := (\mathbf{X}, \mathbf{Y})$

KeyGen:

1. Sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$
2. Sample $\mathbf{X} := \mathbf{A} \leftarrow \mathbb{Z}_q^{(2n \log q) \times n}$
3. Sample $\mathbf{z} \leftarrow \chi^{2n \log q}$ (can use wPRF construction as above).
4. Compute $\mathbf{Y} := \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{z}$
5. $\text{sk} := \mathbf{s}, \text{pk} := (\mathbf{A}, \mathbf{b})$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0, 1\}^n$
2. Compute $\mathbf{c} := \bigoplus_{0 \leq j < n} x_{r_j, j}$
3. Compute $e := m \otimes \left(\bigotimes_{0 \leq j < n} y_{r_j, j} \right)$
4. Output $(\mathbf{c}, e)$

Dec:

1. Output $m' := e \otimes F_k(\mathbf{c})^{-1}$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0, 1\}^{2n \log q}$
2. Compute $\mathbf{c} := \mathbf{r}^T \cdot \mathbf{A}$
3. Compute $e := \lfloor \frac{q}{2} \rfloor \cdot m + \mathbf{r}^T \cdot \mathbf{b}$
4. Output $(\mathbf{c}, e)$

Dec:

1. Output $\mathcal{R}(e - \langle \mathbf{c}, \mathbf{s} \rangle)$

DRAGONS!

Game ₃	$C(m_0, m_1)$
$(X, Y), k \leftarrow \text{KeyGen}()$	$r \leftarrow \{0, 1\}^n$
$b \leftarrow \{0, 1\}$	$c \leftarrow \bigoplus_{0 \leq j < n} X_{r_j, j}$
$b' \leftarrow \mathcal{D}^c$	$e \leftarrow m_b \otimes F_k(c)$
return $b = b'$	return (c, e)

Game ₃	$C(m_0, m_1)$
$(A, b), s \leftarrow \text{KeyGen}()$	$r \leftarrow \{0, 1\}^{2n \log q}$
$b \leftarrow \{0, 1\}$	$c \leftarrow r^T \cdot A$
$b' \leftarrow \mathcal{D}^c$	$e \leftarrow \lfloor \frac{q}{2} \rfloor \cdot m + r^T \cdot b$
return $b = b'$	return (c, e)

- We cannot rely on correctness here because

$$\begin{aligned}
 F_{(k,s)}(\mathbf{a}_0) + F_{(k,s)}(\mathbf{a}_1) &= \langle \mathbf{a}_0, \mathbf{s} \rangle + D_\chi(F'_k(\mathbf{a}_0)) + \langle \mathbf{a}_1, \mathbf{s} \rangle + D_\chi(F'_k(\mathbf{a}_1)) \\
 &\neq \langle \mathbf{a}_0 + \mathbf{a}_1, \mathbf{s} \rangle + D_\chi(F'_k(\mathbf{a}_0 + \mathbf{a}_1)) = F_{(k,s)}(\mathbf{a}_0 + \mathbf{a}_1)
 \end{aligned}$$

- We only have $\mathcal{R}(F_{(k,s)}(\mathbf{a}_0) + F_{(k,s)}(\mathbf{a}_1)) = \mathcal{R}(F_{(k,s)}(\mathbf{a}_0 + \mathbf{a}_1))$; could output $\mathcal{R}(e)$ instead.
- Regev solved this directly by appealing to the LHL on $(A, b) \in \mathbb{Z}_q^{(2n \log q) \times (n+1)}$

Richard Lindner and Chris Peikert. **Better Key Sizes (and Attacks) for LWE-Based Encryption**. In: *CT-RSA 2011*. Ed. by Aggelos Kiayias. Vol. 6558. LNCS. Springer, Berlin, Heidelberg, Feb. 2011, pp. 319–339. DOI: [10.1007/978-3-642-19074-2_21](https://doi.org/10.1007/978-3-642-19074-2_21)

Key Idea: replace LHL with applying LWE again.

KeyGen:

1. Sample $\mathbf{s} \leftarrow \$ \mathbb{Z}_q^k$
2. Sample $\mathbf{X} := \mathbf{A} \leftarrow \$ \mathbb{Z}_q^{(2n \log q) \times n}$
3. Sample $\mathbf{e} \leftarrow \$ \chi^{2n \log q}$.
4. Compute $\mathbf{Y} := \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e}$
5. $\text{sk} := \mathbf{s}, \text{pk} := (\mathbf{A}, \mathbf{b})$

Enc:

1. Sample $\mathbf{r} \leftarrow \$ \{0, 1\}^{2n \log q}$
2. Compute $\mathbf{c} := \mathbf{r}^T \cdot \mathbf{A}$
3. Compute $e := \lfloor \frac{q}{2} \rfloor \cdot m + \mathbf{r}^T \cdot \mathbf{b}$
4. Output $(\mathbf{c}, e)$

Dec:

1. Output $\mathcal{R}(e - \langle \mathbf{c}, \mathbf{s} \rangle)$

KeyGen:

1. Sample $\mathbf{s} \leftarrow \$ \mathbb{Z}_q^k$
2. Sample $\mathbf{X} := \mathbf{A} \leftarrow \$ \mathbb{Z}_q^{n \times n}$
3. Sample $\mathbf{e} \leftarrow \$ \chi^n$.
4. Compute $\mathbf{Y} := \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e}$
5. $\text{sk} := \mathbf{s}, \text{pk} := (\mathbf{A}, \mathbf{b})$

Enc:

1. Sample $\mathbf{r}, \mathbf{d}, \mathbf{d}' \leftarrow \$ \chi^n, \chi^n, \chi$
2. Compute $\mathbf{c} := \mathbf{r}^T \cdot \mathbf{A} + \mathbf{d}$
3. Compute $e := \lfloor \frac{q}{2} \rfloor \cdot m + \mathbf{r}^T \cdot \mathbf{b} + \mathbf{d}'$
4. Output $(\mathbf{c}, e)$

Dec:

1. Output $\mathcal{R}(e - \langle \mathbf{c}, \mathbf{s} \rangle)$

FIN

READ MINICRYPT PRIMITIVES WITH ALGEBRAIC STRUCTURE AND
APPLICATIONS!

- [AMPR23] Navid Alamati, Hart Montgomery, Sikhar Patranabis, and Arnab Roy. **Minicrypt Primitives with Algebraic Structure and Applications**. In: *Journal of Cryptology* 36.1 (Jan. 2023), p. 2. DOI: 10.1007/s00145-022-09442-2.
- [BM84] Manuel Blum and Silvio Micali. **How to Generate Cryptographically Strong Sequences of Pseudorandom Bits**. In: *SIAM Journal on Computing* 13.4 (1984), pp. 850–864.
- [ElG84] Taher ElGamal. **A Public Key Cryptosystem and a Signature Scheme Based on Discrete Logarithms**. In: *CRYPTO'84*. Ed. by G. R. Blakley and David Chaum. Vol. 196. LNCS. Springer, Berlin, Heidelberg, Aug. 1984, pp. 10–18. DOI: 10.1007/3-540-39568-7_2.

REFERENCES II

- [GGM86] Oded Goldreich, Shafi Goldwasser, and Silvio Micali. **How to Construct Random Functions**. In: *Journal of the ACM* 33.4 (Oct. 1986), pp. 792–807. DOI: 10.1145/6490.6503.
- [Lam79] Leslie Lamport. **Constructing Digital Signatures from a One-way Function**. Technical Report SRI-CSL-98. SRI International Computer Science Laboratory, Oct. 1979.
- [LP11] Richard Lindner and Chris Peikert. **Better Key Sizes (and Attacks) for LWE-Based Encryption**. In: *CT-RSA 2011*. Ed. by Aggelos Kiayias. Vol. 6558. LNCS. Springer, Berlin, Heidelberg, Feb. 2011, pp. 319–339. DOI: 10.1007/978-3-642-19074-2_21.
- [LR88] Michael Luby and Charles Rackoff. **How to construct pseudorandom permutations from pseudorandom functions**. In: *SIAM Journal on Computing* 17.2 (1988).

- [Mer79] Ralph Charles Merkle. **Secrecy, authentication, and public key systems.** PhD thesis. Stanford university, 1979.
- [NY89] Moni Naor and Moti Yung. **Universal One-Way Hash Functions and their Cryptographic Applications.** In: *21st ACM STOC*. ACM Press, May 1989, pp. 33–43. DOI: [10.1145/73007.73011](https://doi.org/10.1145/73007.73011).
- [Reg09] Oded Regev. **On lattices, learning with errors, random linear codes, and cryptography.** In: *Journal of the ACM* 56.6 (Sept. 2009), 34:1–34:40. ISSN: 0004-5411 (print), 1557-735X (electronic). DOI: <http://doi.acm.org/10.1145/1568318.1568324>.
- [Rot11] Ron Rothblum. **Homomorphic Encryption: From Private-Key to Public-Key.** In: *TCC 2011*. Ed. by Yuval Ishai. Vol. 6597. LNCS. Springer, Berlin, Heidelberg, Mar. 2011, pp. 219–234. DOI: [10.1007/978-3-642-19571-6_14](https://doi.org/10.1007/978-3-642-19571-6_14).

- [Sch90] Claus-Peter Schnorr. **Efficient Identification and Signatures for Smart Cards**. In: *CRYPTO'89*. Ed. by Gilles Brassard. Vol. 435. LNCS. Springer, New York, Aug. 1990, pp. 239–252. DOI: [10.1007/0-387-34805-0_22](https://doi.org/10.1007/0-387-34805-0_22).